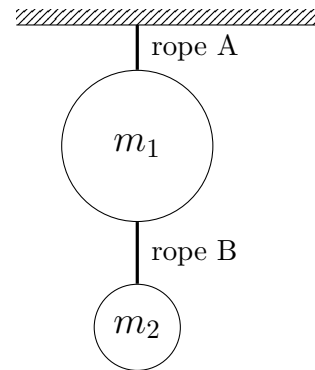

Task #1

Consider a 1400-kg car that experiences 900 N of friction and 1300 N of air drag. How much force must the car's engine exert in order to maintain a constant speed of 45 m/s?

Task #2

Two masses are suspended at rest. Assume that $m_1 = 5.2$ kg and $m_2 = 1.7$ kg.

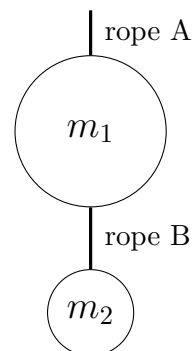
- (a) Draw a free-body diagram for each mass.
- (b) Calculate the tension on each rope.



Task #3

Now you've detached rope A and are accelerating the system upward at a rate of 0.5 m/s^2 , what would be the tensions on both ropes?

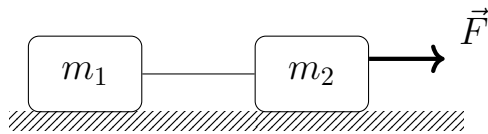
- (a) Draw a free-body diagram for each mass.
- (b) Calculate the tension on each rope.



Task #4

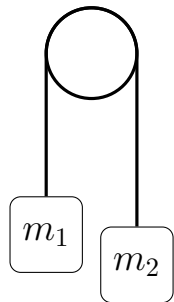
Two boxes have masses $m_1 = 20$ kg and $m_2 = 10$ kg and are sitting on a frictionless surface connected by a massless cord. They are pulled with an applied force of $F = 50$ N.

- (a) Draw a free-body diagram for each mass.
- (b) Calculate the acceleration of the system.
- (c) Calculate the tension in the cord connecting them.



Task #5

Two masses $m_1 = 4$ kg and $m_2 = 2$ kg are attached by a string that hangs over a frictionless pulley. What is (a) the tension on the string and (b) the acceleration of the masses? (This is known as an *Atwood Machine*)



Task #6

We now have what is called a *Modified Atwood Machine* with $m_1 = 4$ kg and $m_2 = 3$ kg. What is (a) the tension on the string and (b) the acceleration of the masses? Again, the surface is frictionless.

